

Comprehensive System of University Management

Integrantes:

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Purpose:

Manage the data ecosystem of an academic institution through a robust architecture.

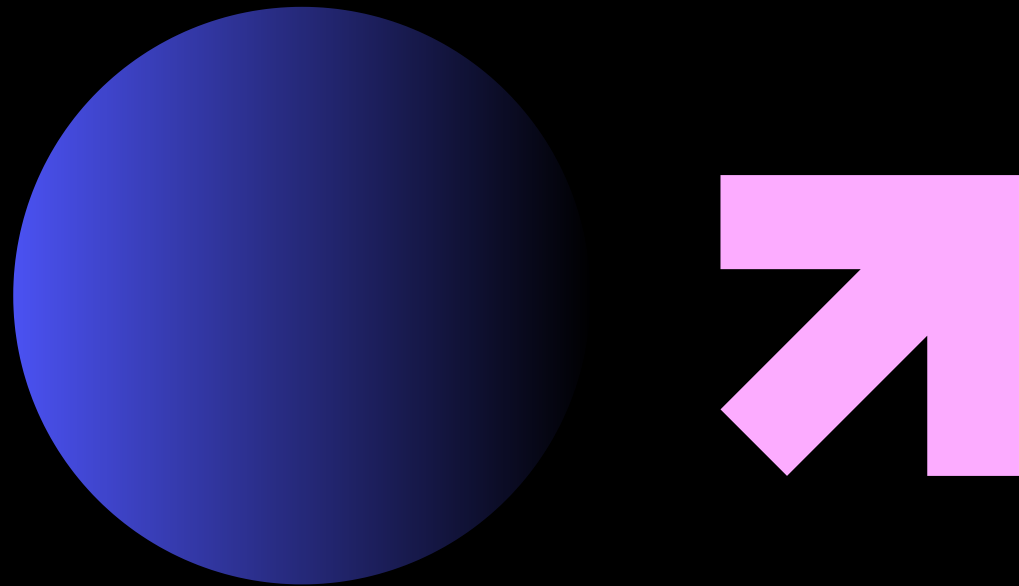
Main goals:

- Centralize information from the university community
- Control academic processes (enrollments, campus/faculty changes)
- Automate the generation of official documents
- Maintain strict referential integrity

preview:

6 Use Cases Developed

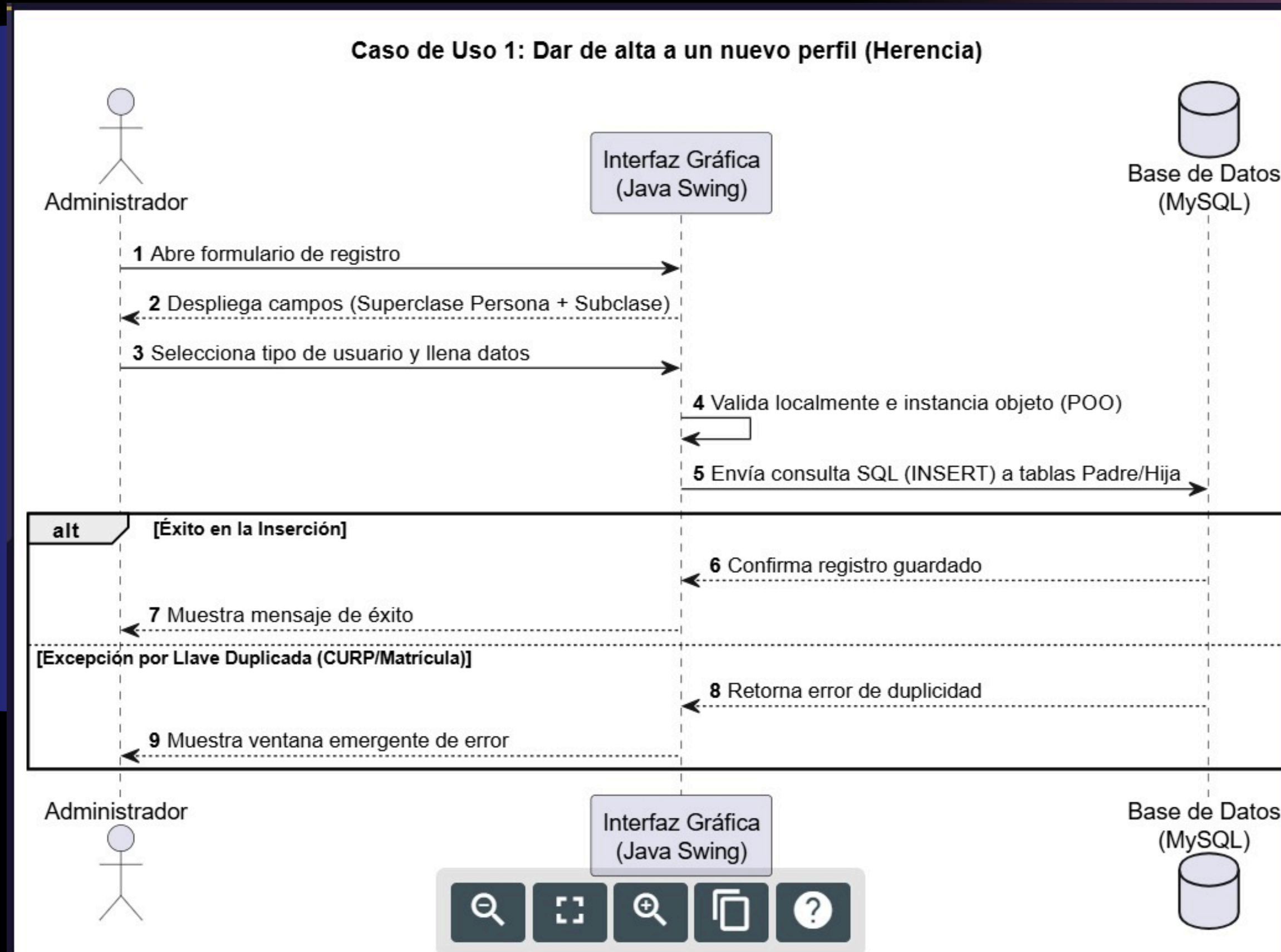
FUNCTIONAL REQUIREMENTS AND TECHNICAL BASIS



- Technologies: Java (Swing), PlantUML for diagrams
- GitHub Repository: Structure organized by use case (.puml)

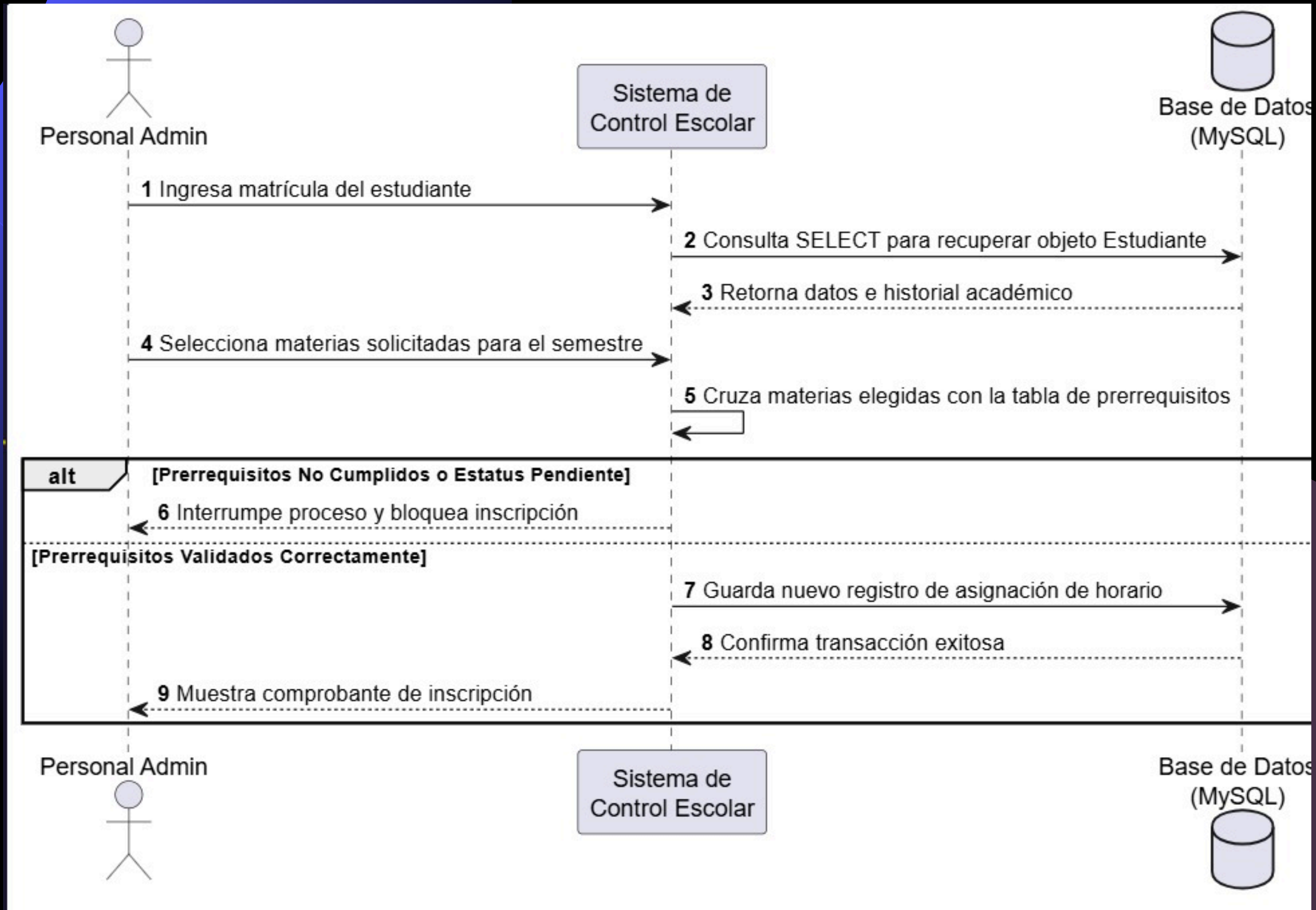
Module	Key feature
User Mangement	Hierarchical profiles with Inheritance: Person → Student / Worker / Administrative
Module Acdemic	Registrations, prerequisite validation, faculty changes
Dynamic Responses	Automatic generation of documents (.RTF) ready for printing
Hybrid Resistance	Local backup for server connection failures

USE CASE 1: PROFILE REGISTRATION (INHERITANCE)



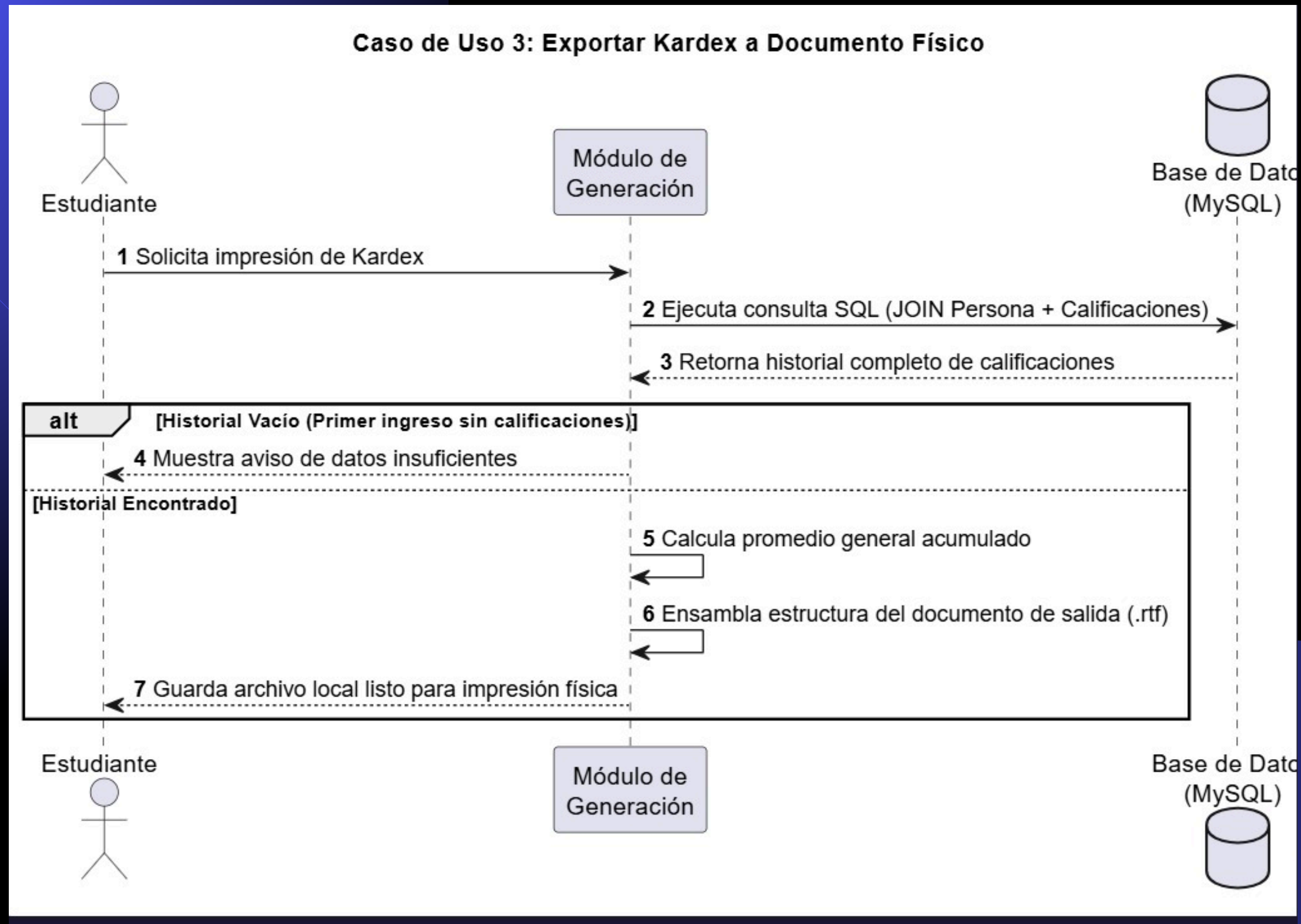
USE CASE 2: ENROLLMENT AND PREREQUISITE VALIDATION

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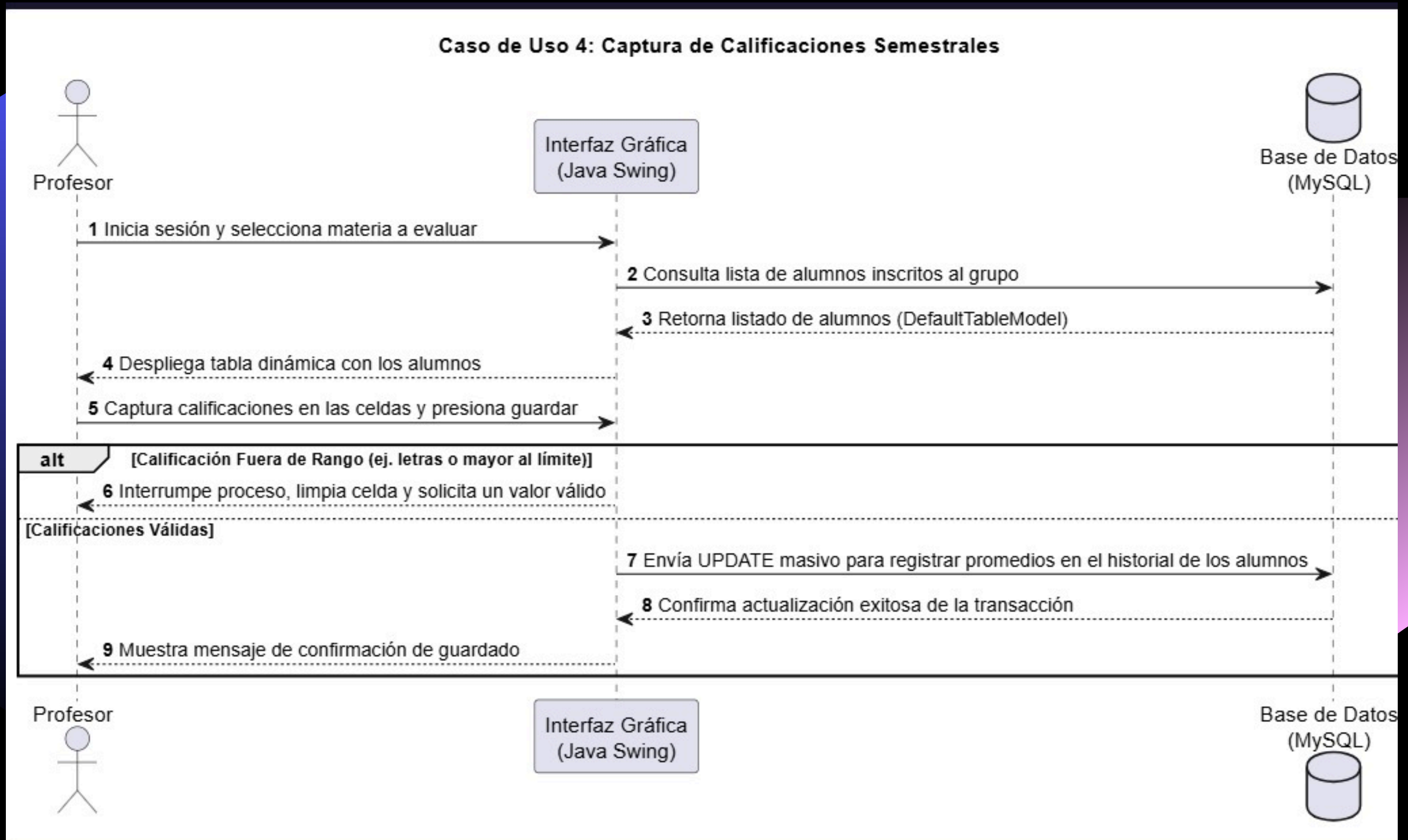


USE CASE 3: EXPORT KARDEX TO DOCUMENT

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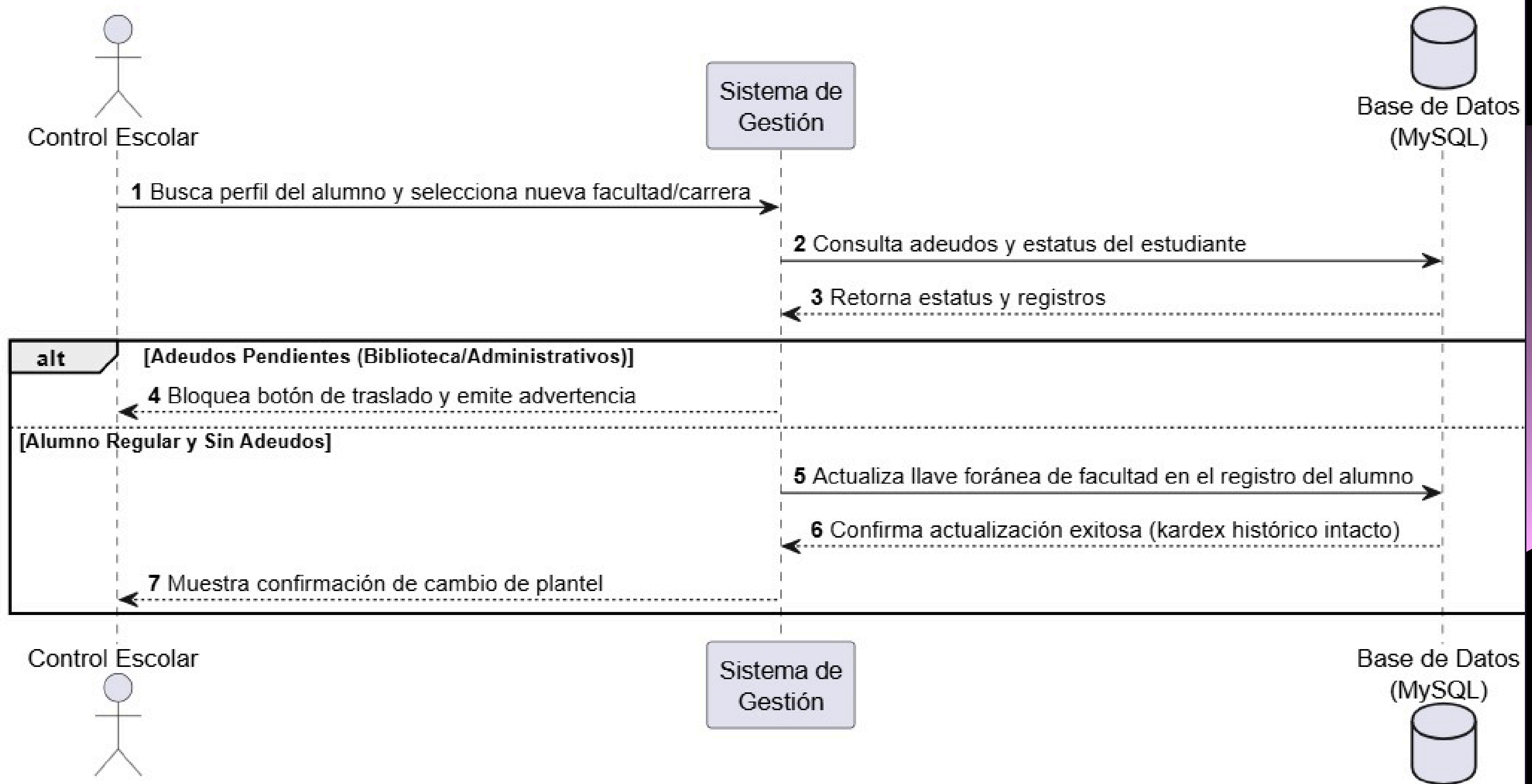
USE CASES 4: GRADES



USE CASES 5: CHANGING FACULTY

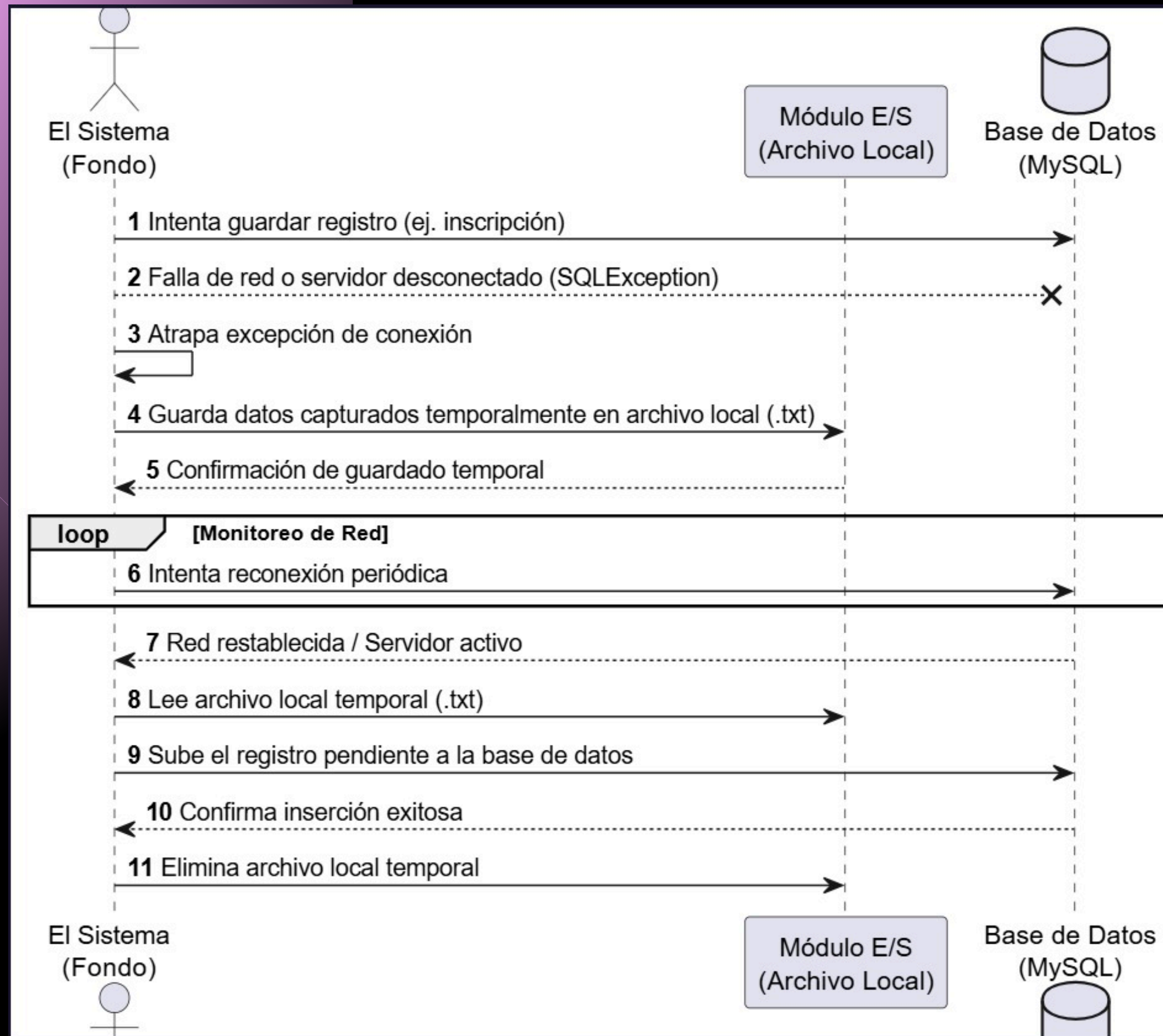
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Caso de Uso 5: Trámite de Cambio de Facultad



USE CASE 6: EMERGENCY BACKUP

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CONCLUSIONS



This project shows the analysis and design of a real system through its use cases and diagrams. Through inheritance, business rules, and validations, we define how each process in the institution should work: from registering people to generating reports and protecting information. Everything is organized so that when it's time to program it, a neat, safe, and reliable system is built. In short: we plan clearly what is going to be done before actually doing it.

Thank You